



Netramind

STANDARD OPERATING PROCEDURE

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Title: Audit Trail Review**1. PURPOSE**

This Standard Operating Procedure (SOP) describes the requirement and process for periodic review of audit trails generated by GxP regulated computerized systems to assure appropriate traceability of data transformation relevant to data integrity in the conduct of clinical trials, and may be supplemented by separate procedures for commercial or post-market systems, as applicable.

2. SCOPE

This SOP applies to audit trail review of GxP computerized systems used by Netramind to generate, process, or report clinical trial data.

Netramind may delegate tasks from this SOP to a service provider if Netramind has determined their SOPs are acceptable for use. However, Netramind retains overall responsibility for those delegated activities (ICH E6[R3], Section II.10.1).

3. REFERENCES

- European Medicines Agency (EMA): EudraLex Volume 4, Part 1: Basic Requirements for Medicinal Products; Annex 11: Computerized Systems
- European Medicines Agency (EMA): Guideline on computerized systems and electronic data in clinical trials
- US Food and Drug Administration (FDA) Guidance, *Part 11, Electronic Records; Electronic Signatures - Scope and Application* (August 2003)
- US Food and Drug Administration (FDA): 21 CFR Part 11; 21 CFR parts 210 and 211 and 820
- International Conference on Harmonization (ICH) Q7: Good Manufacturing Practice Guide for Active Pharmaceutical Ingredients; November 2000
- International Conference on Harmonization (ICH) E6 (R3) Good Clinical Practice, September 2025
- [SOP-016 Data Integrity](#)
- [SOP-015 21CFR Part 11 Assessment](#)

4. DEFINITIONS

- 4.1. **Audit Trail**— Metadata records that allow the appropriate evaluation of the course of events by capturing details on actions (manual or automated) performed relating to information and data collection and, where applicable, to activities in computerized systems. The audit trail should show activities, initial entry, and changes to data fields or records, by whom, when and, where applicable, why. In computerized systems, the audit trail should be secure, computer-generated, and time stamped.
- 4.2. **Computerized Systems Validation (CSV)**— A process of establishing and documenting that the specified requirements of a computerized system can be consistently fulfilled from



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design until decommissioning of the system or transition to a new system. The approach to validation should be based on a risk assessment that takes into consideration the intended use of the system and the potential of the system to affect trial participant protection and the reliability of trial results

- 4.3. Data Acquisition Tool (DAT).—A paper or electronic tool designed to collect data and associated metadata from a data originator in a clinical trial according to the protocol and to report the data to the sponsor. The data originator may be a human (e.g., the participant or trial staff), a machine (e.g., wearables and sensors), or a computer system from which the electronic transfer of data from one system to another has been undertaken (e.g., extraction of data from an electronic health record or laboratory system). Examples of DATs include but are not limited to CRFs, interactive response technologies, clinical outcome assessments, including patient-reported outcomes (PROs) and wearable devices, irrespective of the media used.
- 4.4. Data Integrity.—Data integrity includes the degree to which data fulfill key criteria of being attributable, legible, contemporaneous, original, accurate, complete, secure and reliable such that data are fit for purpose.
- 4.5. Human Readable Format.—Audit trail data that can be interpreted and reviewed by a user without requiring specialized system-level access.
- 4.6. Metadata.— The contextual information required to understand a given data element. Metadata is structured information that describes, explains, or otherwise makes it easier to retrieve, use, or manage data to allow for the appropriate evaluation of the trial conduct.
- 4.7. Raw Data.—Factual recorded information obtained as a result of original observations or activities, which has not been copied or transcribed. This term is synonymous with original data.
- 4.8. Signature.—A unique mark, symbol, or entry executed, adopted, or authorized by an individual, in accordance with applicable regulatory requirements and/or practice to show expression of will and allow authentication of the signatory (i.e., establish a high degree of certainty that a record was signed by the claimed signatory). A signature may be physical or electronic
- 4.9. Source Records.— Original documents or data (which includes relevant metadata) or certified copies of the original documents or data, irrespective of the media used. This may include trial participants' medical/health records/notes/charts; data provided/entered by trial participants (e.g., electronic patient-reported outcomes (ePROs)); healthcare professionals' records from pharmacies, laboratories, and other facilities involved in the clinical trial; and data from automated instruments, such as wearables and sensors.
- 4.10. Subject Matter Expert (SME).—An individual designated by management as an expert in a particular area based on attributes such as education, training, and experience.

5. RESPONSIBILITIES

- 5.1. Business Process Owner (BPO)



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- 5.1.1. Identify the regulatory and business requirements that the computerized system must comply with based on risk assessments.
- 5.1.2. Review and approve audit trail specifications.
- 5.1.3. Resolve audit trail issues with IT Quality Assurance (QA).
- 5.1.4. Ensure that the audit trail function is always enabled and not modifiable in GxP-regulated computer systems.
- 5.1.5. Coordinate with IT and/or System Owner to designate qualified audit trail reviewers.
- 5.1.6. Approve Audit Trail Reports.
- 5.2. Designated Audit Trail Reviewer (may include System Administrator or trained personnel)
 - 5.2.1. Performs audit trail reviews.
- 5.3. System Owner (SO)
 - 5.3.1. Ensure system users have requisite training for system use.
 - 5.3.2. Approve system access to authorized system users.
 - 5.3.3. Approve Audit Trail Reports.
 - 5.3.4. Coordinate with the Business Process Owner to designate qualified audit trail reviewers.
- 5.4. Quality Assurance (QA)
 - 5.4.1. Provides Audit Trail oversight.
 - 5.4.2. Approves Audit Trail Report.

6. PROCEDURE

- 6.1. Identifying Regulatory and Business Requirements for Computerized Systems
 - 6.1.1. The BPO conducts a risk assessment of the computerized system and associated platforms to ensure it meets the intended use for business needs.
 - 6.1.2. For systems which generate, store, report on data collected for GxP purpose, the BPO ensures that the system(s) complies with applicable regulatory requirements, in accordance with **SOP-015 21CFR Part 11** and **SOP-016 Data Integrity**.
 - 6.1.3. Considerations during the computerized system risk assessment should include:
 - 6.1.3.1. Ensuring that metadata is captured with acquired data
 - 6.1.3.2. Automated data validation checks
 - 6.1.3.3. User account management
 - 6.1.3.4. Traceability of data changes
 - 6.1.3.5. Integrated workflows
 - 6.1.3.6. Automated capture of date and time during data capture and transfers
 - 6.1.3.7. Periodic review of trial-specific data, Audit Trails, and other metadata



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6.1.3.8. Documentation of change controls

6.1.4. The BPO documents and communicates the risk assessment results with the IT System Owner(s) and System Administrator(s).

6.2. Establishing Audit Trail Items for a Computerized System

6.2.1. The Business Process Owner (BPO), System Owner, and QA jointly define audit trail scope, review items, and frequency, based on the risk category assigned to each computerized system.

6.2.2. The extent and nature of periodic review of audit trails should be risk-based:

6.2.2.1. Review should be adapted to meet the trial design, phase, and complexity of the data, as defined by critical data elements and processes.

6.2.2.2. Review should focus on critical systems that process critical data elements and the relevant metadata.

6.2.3. Audit trails and relevant metadata must be recorded in such a way that is decipherable, changes should be supported by source data around the time of the original entry.

6.2.4. The BPO, System Owner, and QA use the results from the risk assessment to establish audit review items, including the review frequency, based on the risk category assigned to each computerized system.

6.2.4.1. System or Process Owner determines audit trails needs based on system and study specific requirements.

6.2.4.2. QA defines the requirements for audit trail transactions to comply with ICH standards.

6.2.4.3. QA determines the requirements for audit trail reviews, including relevant metadata, and frequency of reviews.

6.2.5. End-users assigned to perform periodic reviews of each system and/or each clinical trial are trained and understand the requirements for reviewing and documenting the results.

6.3. Conducting an Audit Trail Review

6.3.1. Audit trail reviews should focus on identifying unauthorized changes, unexpected data modifications, and system misuse.

6.3.2. The BPO and SO designate a qualified audit trail reviewer to conduct periodic Audit Trail reviews, based on each computerized system and for each relevant clinical trial.

6.3.3. The designated System Administrator and/or Audit Trail Reviewer schedules an Audit Trail periodic review based on the frequency defined in the risk assessment.



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6.3.4. The following requirements and methods may be utilized:

- 6.3.4.1. Ensure that audit trails are available in Human Readable Format
- 6.3.4.2. Audit trail reviews are conducted either manually or using software tools such as SQL queries, tag filters, etc. based on the established system requirements.
- 6.3.4.3. Audit Trail review should evaluate activities, initial entries, and changes to data fields or records, by whom, when and, where applicable, why.
- 6.3.4.4. Additional information to clarify anomalies in the audit trail may be required, such as study team list, data handling guidelines, etc. which the audit trail reviewer should be informed prior to the review. The reviewer can also request additional clarifications from the SO and/or study SMEs.
- 6.3.4.5. Document each finding with sufficient details to reference the discrepancies.
- 6.3.4.6. Verification that changes to critical data are appropriately justified and documented.

6.3.5. The review goal is to ensure record integrity through examination of data security. The Audit Trail review should, at minimum, consist of the following:

Event Types	Actions
Account management	Monitor changes to user and computer accounts and groups
Detailed tracking of applications	Ensure that major configuration changes/ revisions are supported by approved quality records
Intrusion attempts	Review login audit trails to identify a large number of consecutive and repeated failed login attempts with the same User ID and incorrect password
Disabling of the audit trail feature (if applicable)	Number of times that the audit trail feature was turned on/off, accompanied by the date and time of each activity

6.4. Audit Trail Periodic Review Report

- 6.4.1. Documentation may be maintained in a structured report, log, or tracking system, provided traceability is maintained.
- 6.4.2. The System Administrator and/or Reviewer captures the results of the Audit Trail periodic review in a report (Appendix 1).
- 6.4.3. Identified issues and/or unusual incidents are escalated to appropriate Functional Area management personnel and QA for resolution.



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- 6.4.3.1. Each observation is assigned a unique Audit Trail Review (ATD) ID number (i.e., ATD-001).
- 6.4.3.2. Additional details, e.g., record numbers, date of event, captured image of record can be supplemented to better assess the finding(s).
- 6.4.3.3. If systematic observations are noted, the report should include recommendations for further investigation or increased review scope (i.e., frequency, types of data, sampling size) to fully assess the root cause of the findings.
- 6.4.4. The System Administrator and/or Reviewer distribute the completed Audit Trail Review Report to the BPO, System Owner, and QA for review and approval. Additional stakeholders may also be provided with the report for review, e.g., Functional Area Management, SMEs, etc.
- 6.4.5. The BPO, System Owner, and QA review and approve the completed Audit Trail Report.
 - 6.4.5.1. If additional review or corrective actions are required, the BPO, SO and QA determine whether further investigation, corrective action, or escalation is required.
 - 6.4.5.2. The recommendations are communicated to relevant SO, SMEs, and/or Functional Area Management for further resolution.
- 6.4.6. The approved Audit Trail Report is filed and archived in the relevant system accessible to the BPO, System Owner, and QA.
- 6.4.7. A copy of the Audit Trail Report can be filed in the study TMF, if the report was generated for a study-specific system and/or process.
- 6.5. Compliance with Periodic Reviews
 - 6.5.1. The BPO, System Owner and QA maintains a record of scheduled and completed audit trail reviews for each computer system and/or study and are responsible for adhering to the scheduled periodic review as required in the initial risks assessment, e.g., annually.
 - 6.5.2. If the periodic review frequency changes, due to system status, study closure, etc. the BPO, System Owner and QA ensure the changes are captured on the tracker, along with the rationale for updating the periodic review frequency or requirements.

7. ATTACHMENTS

- [Appendix 1: Audit Trail Review Report Template](#)



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8. REVISION HISTORY

Version Number	Date	Summary of Change(s)
01	01-MAY-2026	New Document
[X.0]	[DD-MMM-YYYY]	[Insert a summary of the changes]



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Appendix 1: Audit Trail Review Report Template

System Name:	Review Date:
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ID	Observation Details	Observation Follow-up/ Resolution	Responsible Person

Audit Trail Review Report Approval

	Print Name	Signature	Date
Performed By:			
BPO Approval:			
SO Approval:			
IT QA Approval:			



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SIGNATURE PAGE

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APPROVER

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